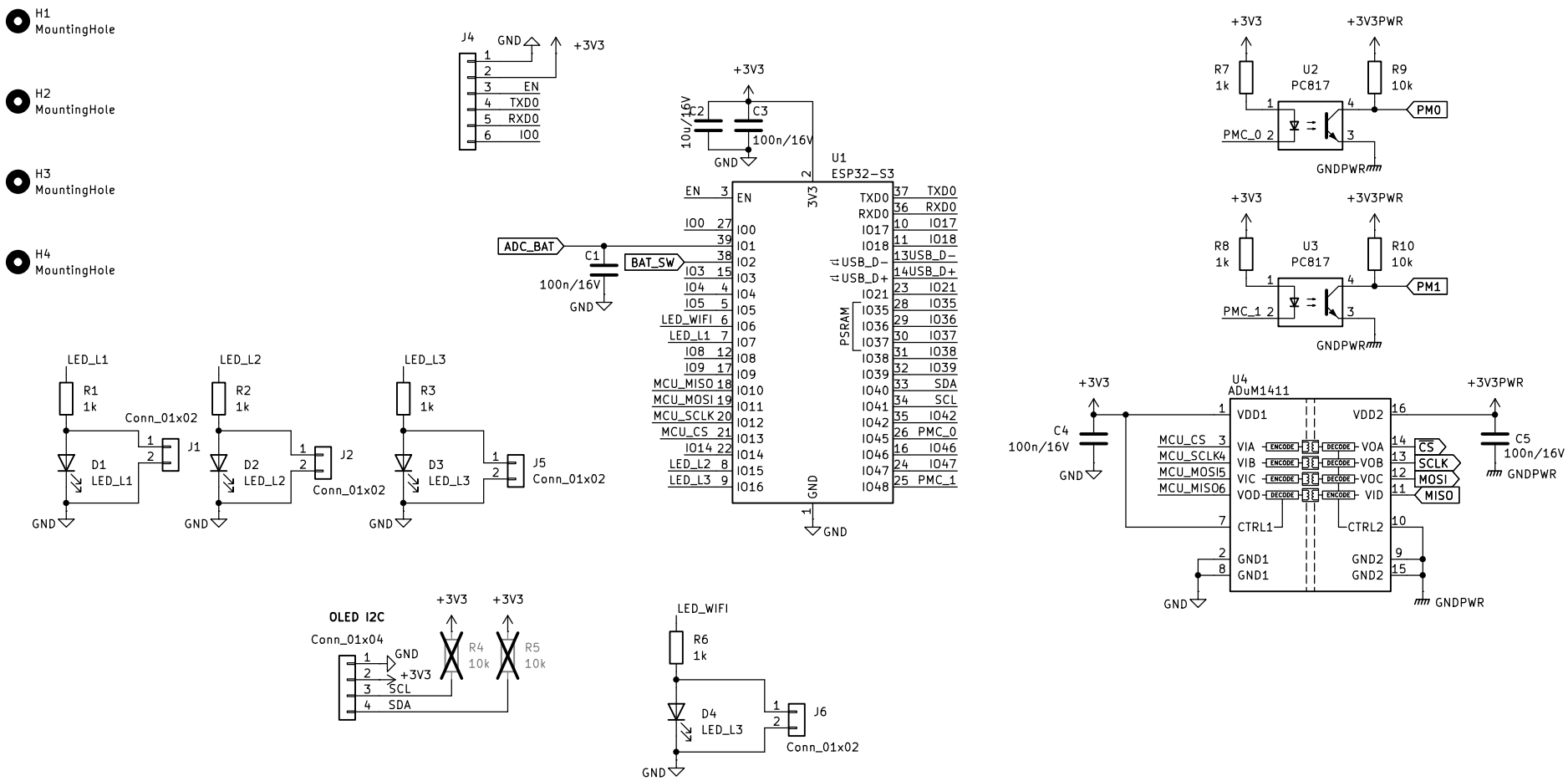
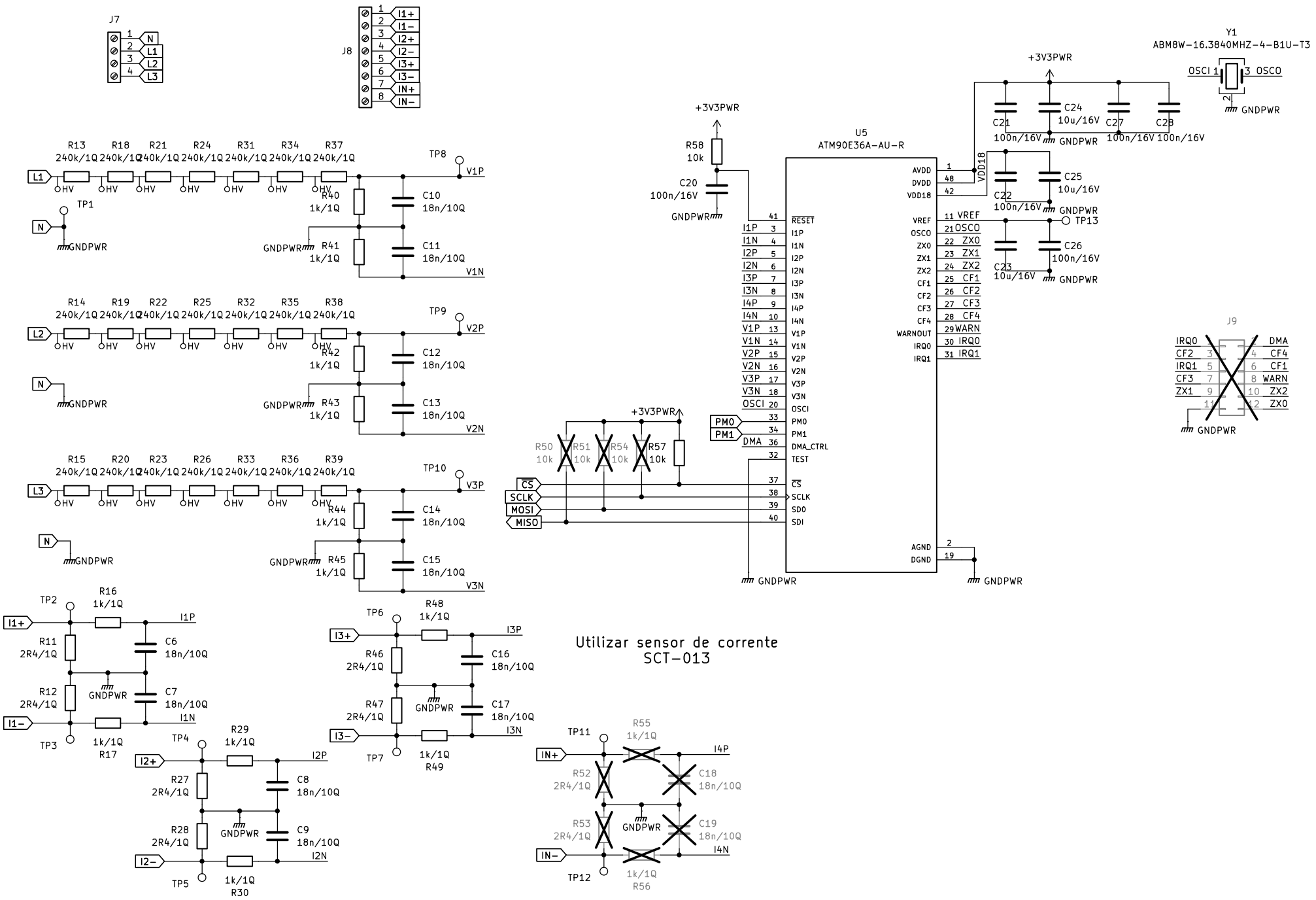
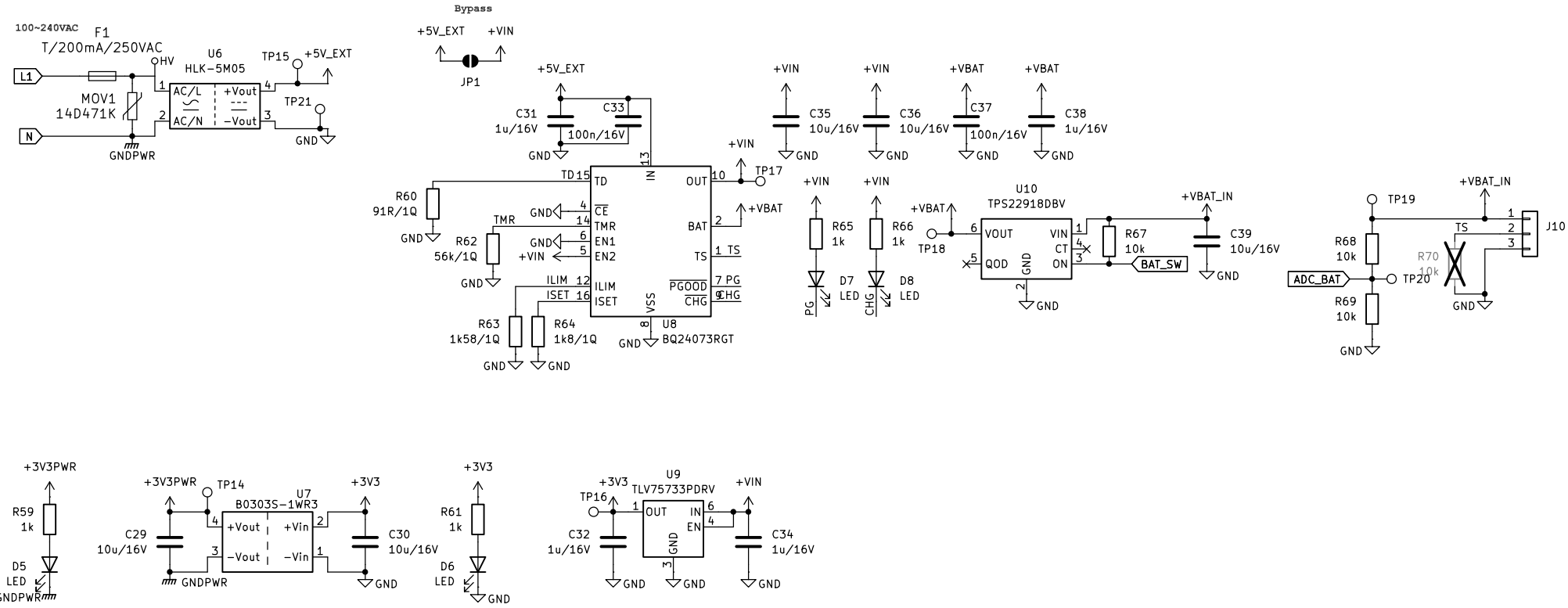

H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
J4
GND
+3V3
EN
TXD0
RXD0
IO0
+3V3
C2
10u/16V
C3
100n/16V
GND
U1
ESP32-S3
ADC_BAT
C1
100n/16V
BAT_SW
LED_WIFI
LED_L1
LED_L2
LED_L3
MCU_MISO
MCU_MOSI
MCU_SCLK
MCU_CS
PSRAM
USB_D-
USB_D+
PMC_0
PMC_1
+3V3
+3V3PWR
R7
1k
U2
PC817
R9
10k
PM0
PMC_0
GNDPWR
R8
1k
U3
PC817
R10
10k
PM1
PMC_1
GNDPWR
LED_L1
R1
1k
Conn_01x02
J1
D1
LED_L1
LED_L2
R2
1k
J2
D2
LED_L2
Conn_01x02
LED_L3
R3
1k
J5
D3
LED_L3
Conn_01x02
GND
OLED I2C
Conn_01x04
GND
+3V3
SCL
SDA
R4
10k
R5
10k
LED_WIFI
R6
1k
D4
LED_L3
J6
Conn_01x02
GND
+3V3
C4
100n/16V
GND
U4
ADuM1411
MCU_CS
MCU_SCLK
MCU_MOSI
MCU_MISO
VDD1
VDD2
VIA
VIB
VIC
VOD
VOA
VOB
VOC
VID
ENCODE
DECODE
CTRL1
CTRL2
GND1
GND2
CS
SCLK
MOSI
MISO
+3V3PWR
C5
100n/16V
GNDPWR
GND
GNDPWR


J7
1 N
2 L1
3 L2
4 L3
J8
1 I1+
2 I1−
3 I2+
4 I2−
5 I3+
6 I3−
7 IN+
8 IN−
Y1
ABM8W-16.3840MHZ-4-B1U-T3
OSCI 1
3 OSCO
2
GNDPWR
+3V3PWR
C21 100n/16V
C24 10u/16V
C27 100n/16V
C28 100n/16V
GNDPWR
C22 100n/16V
C25 10u/16V
C23 10u/16V
C26 100n/16V
VDD18
R58 10k
C20 100n/16V
U5
ATM90E36A-AU-R
AVDD 1
DVDD 48
VDD18 42
VREF 11
OSCO 21
ZX0 22
ZX1 23
ZX2 24
CF1 25
CF2 26
CF3 27
CF4 28
WARNOUT 29
IRQ0 30
IRQ1 31
RESET 41
I1P 3
I1N 4
I2P 5
I2N 6
I3P 7
I3N 8
I4P 9
I4N 10
V1P 13
V1N 14
V2P 15
V2N 16
V3P 17
V3N 18
OSCI 20
PM0 33
PM1 34
DMA 36
TEST 32
CS 37
SCLK 38
SDO 39
SDI 40
AGND 2
DGND 19
TP13
J9
IRQ0 DMA
CF2 CF4
IRQ1 CF1
CF3 WARN
ZX1 ZX2
ZX0
R13 R18 R21 R24 R31 R34 R37
240k/1Q
L1
HV
TP1
N
GNDPWR
TP8
V1P
R40 1k/1Q
R41 1k/1Q
C10 18n/10Q
C11 18n/10Q
V1N
R14 R19 R22 R25 R32 R35 R38
240k/1Q
L2
TP9
V2P
R42 1k/1Q
R43 1k/1Q
C12 18n/10Q
C13 18n/10Q
V2N
R15 R20 R23 R26 R33 R36 R39
240k/1Q
L3
TP10
V3P
R44 1k/1Q
R45 1k/1Q
C14 18n/10Q
C15 18n/10Q
V3N
+3V3PWR
R50 10k
R51 10k
R54 10k
R57 10k
CS
SCLK
MOSI
MISO
TP2
I1+
R16 1k/1Q
I1P
R11 2R4/1Q
C6 18n/10Q
R12 2R4/1Q
C7 18n/10Q
I1−
I1N
TP3
1k/1Q R17
TP6
I3+
R48 1k/1Q
I3P
R46 2R4/1Q
C16 18n/10Q
R47 2R4/1Q
C17 18n/10Q
I3−
I3N
TP7
1k/1Q R49
TP4
I2+
R29 1k/1Q
I2P
R27 2R4/1Q
C8 18n/10Q
R28 2R4/1Q
C9 18n/10Q
I2−
I2N
TP5
1k/1Q R30
Utilizar sensor de corrente
SCT−013
TP11
IN+
R55 1k/1Q
I4P
R52 2R4/1Q
C18 18n/10Q
R53 2R4/1Q
C19 18n/10Q
IN−
I4N
TP12
1k/1Q R56


100~240VAC
F1
T/200mA/250VAC
HV
L1
N
MOV1
14D471K
GNDPWR
U6
HLK-5M05
AC/L
AC/N
+Vout
-Vout
TP15
+5V_EXT
TP21
GND
Bypass
+5V_EXT
+VIN
JP1
C31
1u/16V
C33
100n/16V
GND
TD15
TD
IN
OUT
TP17
R60
91R/1Q
TMR
R62
56k/1Q
CE
EN1
EN2
BAT
TS
ILIM
ISET
PGOOD
CHG
VSS
R63
1k58/1Q
R64
1k8/1Q
U8
BQ24073RGT
C35
10u/16V
C36
10u/16V
C37
100n/16V
C38
1u/16V
+VBAT
R65
1k
R66
1k
D7
LED
D8
PG
CHG
U10
TPS22918DBV
TP18
VOUT
VIN
CT
QOD
ON
R67
10k
BAT_SW
C39
10u/16V
+VBAT_IN
TP19
J10
R68
10k
R69
10k
ADC_BAT
TP20
TS
R70
10k
+3V3PWR
R59
1k
D5
LED
TP14
B0303S-1WR3
+Vin
-Vin
C29
10u/16V
C30
10u/16V
+3V3
R61
1k
D6
U9
TLV75733PDRV
TP16
C32
1u/16V
C34
1u/16V
EN